

AI / ML Engineer Banne Ka Roadmap

Step-by-step Hindi learning plan — Dikshant Aggarwal ke resume se 100% matched

Base: "Complete Generative AI Course with LangChain and Huggingface" (Udemy) + free Hindi YouTube playlists

Tumhare resume me 6 bade area hain: Python, Machine Learning, Deep Learning, Generative AI, Computer Vision aur MLOps/Web. Tumhara Udemy GenAI course poora **Step 4 (Generative AI)** cover kar deta hai — yahi resume ki sabse strong jagah hai. Baaki steps free Hindi playlists se complete honge. Sabse zaroori **Step 5 (Computer Vision)** aur **Step 6 (MLOps)** hain — yahi tumhare resume ke missing projects ko sach karenge.

Bottom line: Yeh roadmap complete karne ke baad tum resume ke **saare 5 projects** khud bana paoge — aur ek full "AI/ML Engineer" profile ready ho jayegi.

Resume Skills -> Roadmap Coverage (100%)

Resume Skill / Area	Step	Hindi Resource	Status
Python (+ C/C++, Java basics)	1	CodeWithHarry Python	Revise
Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib	2	CampusX 100 Days of ML	Naya
PyTorch, CNNs, Transfer Learning, Model Training	3	CampusX Deep Learning (PyTorch)	Naya
OpenAI/Gemini API, LangChain, LangGraph, RAG, FAISS, HuggingFace, Prompt Eng.	4	Udemy course + CampusX GenAI	Course me - done
Multi-agent workflows, AI automation, Streamlit, Colab, AWS	4	Udemy course (CrewAI, deploy, AWS bonus)	Course me - done
Claude Code, custom skills and plugins	4+	Anthropic docs + practice	Khud practice
OpenCV, YOLOv8, MediaPipe, Object Detection, Image Classification, Roboflow, DeepFace	5	OpenCV + YOLOv8 playlists	GAP - Naya
FastAPI, Docker, MLflow, Git, REST APIs, Deployment	6	CampusX FastAPI + MLOps	GAP - Naya
MySQL/SQL	7	Apna College SQL + Udemy SQL	Mostly done
JS/HTML/CSS, React, Node.js, WebSocket, Firebase	7	CodeWithHarry Web	Optional

Step-by-step Plan (with Hindi playlist links)

STEP 1 Python Foundation		Foundation
Kya seekhoge:	Python (variables, loops, functions, OOP, file handling) - sab kuch ki base.	
Playlist (Hindi):	CodeWithHarry - Python Tutorials in Hindi	
Kaam aayega:	Saare projects ki neev.	
Time (approx):	1-2 hafte (agar already aata hai to revise)	

STEP 2 Machine Learning

Foundation

- Kya seekhoge:** Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, EDA, feature engineering, model evaluation.
- Playlist (Hindi):** [CampusX - 100 Days of Machine Learning](#)
- Kaam aayega:** Project 4 (Production ML API with XGBoost).
- Time (approx):** 4-6 hafte

STEP 3 Deep Learning + PyTorch

Foundation

- Kya seekhoge:** PyTorch, neural networks, CNNs, transfer learning, training pipeline, GPU training.
- Playlist (Hindi):** [CampusX - Practical Deep Learning using PyTorch](#)
- Kaam aayega:** Computer Vision aur fine-tuning ki base.
- Time (approx):** 3-4 hafte

STEP 4 Generative AI and LLMs

Tumhare Udemy course me -
DONE

- Kya seekhoge:** LangChain, LangGraph, RAG, FAISS, OpenAI/Gemini, HuggingFace, fine-tuning, CrewAI agents, prompt engineering.
- Playlist (Hindi):** **Primary:** tumhara **Udemy GenAI course**. **Free Hindi revision:** [CampusX - Generative AI using LangChain](#)
- Kaam aayega:** Project 1 (AI Agent) and Project 2 (RAG).
- Time (approx):** Already chal raha hai

STEP 5 Computer Vision

Resume GAP - yahan focus

- Kya seekhoge:** OpenCV, YOLOv8, custom dataset (Roboflow), object detection, image classification, real-time inference, DeepFace.
- Playlist (Hindi):** [OpenCV Projects \(Hindi/Eng\)](#)
[YOLOv8 Full Tutorial \(Object Detection\)](#)
- Kaam aayega:** Project 3 (Object Detection) and Project 5 (Face Recognition).
- Time (approx):** 4-5 hafte

STEP 6 MLOps and Deployment

Resume GAP - yahan focus

- Kya seekhoge:** FastAPI, Docker, MLflow, Git, REST APIs, model deployment, auto-retraining, AWS.
- Playlist (Hindi):** [CampusX - FastAPI for ML \(FastAPI + Docker + AWS\)](#)
[CampusX - MLflow / End-to-End MLOps playlists](#)
- Kaam aayega:** Saare projects ko production me deploy karna.
- Time (approx):** 3-4 hafte

STEP 7 Web / Backend

Optional (full-stack ke liye)

Kya seekhoge: JavaScript, HTML, CSS, React, Node.js, WebSocket, Firebase, MySQL.

Playlist (Hindi): [CodeWithHarry - React in Hindi](#)
[Apna College - SQL/MySQL One Shot](#) | [CodeWithHarry - Node/Firebase](#)

Kaam aayega: Project 2 (React UI) and Project 5 (Firebase dashboard).

Time (approx): 3-4 hafte

Tumhare 5 Resume Projects -> Kaunse Steps

Resume Project	Steps zaroori
Autonomous AI Agent (LangGraph, OpenAI, FastAPI, Redis)	Step 4 + Step 6
Multimodal RAG System (LangChain, GPT-4o, FAISS, React)	Step 4 + Step 5 + Step 7
Real-Time Object Detection (YOLOv8, OpenCV, FastAPI, Docker)	Step 5 + Step 6
Production ML API + Auto-Retraining (XGBoost, FastAPI, Docker, MLflow)	Step 2 + Step 6
AI Face Recognition (DeepFace, OpenCV, Firebase, FastAPI)	Step 5 + Step 6 + Step 7

Padhne ka tareeka (tips)

Roz 1.5-2 ghante consistent do. Har step ke baad ek chhota project khud banao (theory + practice). Sab kuch GitHub par daalo - yahi resume ka proof banta hai. Step 4 already chal raha hai, isliye saath-saath Step 5 (CV) shuru kar do - yahi sabse bada gap hai.

Sources: CampusX, CodeWithHarry, Apna College (YouTube). Note: Computer Vision aur MLOps me kuch content English/Hindi mix hai - 100% pure Hindi thoda kam available hai.